PRAMAAN प्रमाण

Proof, not a verdict.

Face liveness and presentation-attack detection for the MOSIP Registration Client, desktop and Android, that leaves evidence a server can re-check.

MOSIP Decode 2026 · Problem Statement 04

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A working engine, not yet measured on real attacks

The gap nobody names

The preview is unsigned. The capture is signed. Nothing links them.



- **Capture swap:** whoever controls the preview can pass liveness with one face and enrol another.
- **Android:** the client never sees preview frames at all; the device app shows the stream and returns only the signed capture.
- **Injected video** (OBS, deepfake pipelines) is outside ISO/IEC 30107. **Insiders** can log in with a supervisor's photo.
- **No audit trail:** a client-side pass/fail cannot be re-checked by the server later or re-scored with a better model.

Our answer: two interfaces, seven questions

The problem statement asks to define the frame-streaming interface and the liveness/PAD interface. We did, and bound them together.

FaceFrameSource
frames *and* the signed capture from the same object, so the capture can be bound to what was watched

LivenessEngine
frames in; states, guidance and a decision out; never a score on the screen; fails closed

1	2	3	4	5	6	7
Real sensor? source integrity	One usable face quality gate	Looks alive? passive, asks nothing	Attack marks? print, screen, flat	Fusion accept / challenge / reject	Live answer? random challenge	Same face? binding + signed evidence

- **Mode C (desktop):** the engine runs in the client, on the unsigned M-JPEG stream (3 fps guaranteed, so every cue is gated on the measured frame rate).
- **Mode D (Android):** the engine ships as a library inside the SBI device app, where the frames actually are.

What is different

Each idea is built and tested. None is claimed beyond what the claim sheet allows.

Capture binding

Four locks before any accept: nonce echoed in the device-signed capture, frame hash chain, identity continuity, time. Works on **unmodified SBI**.

Evidence, not a verdict

A signed JWS in the face BIR. One command re-verifies it; a newer model can re-score stored crops. Merkle transparency log.

Cues that fail differently

Passive, moiré, flash reflection (20.7-bit random colour code), scene, stereo. Each abstains when the station lacks the hardware.

Injection defence

Device identity chain, OS camera registrations (OBS found via DirectShow), JPEG stream fingerprint. Not a renamed-driver defence.

Inclusion by design

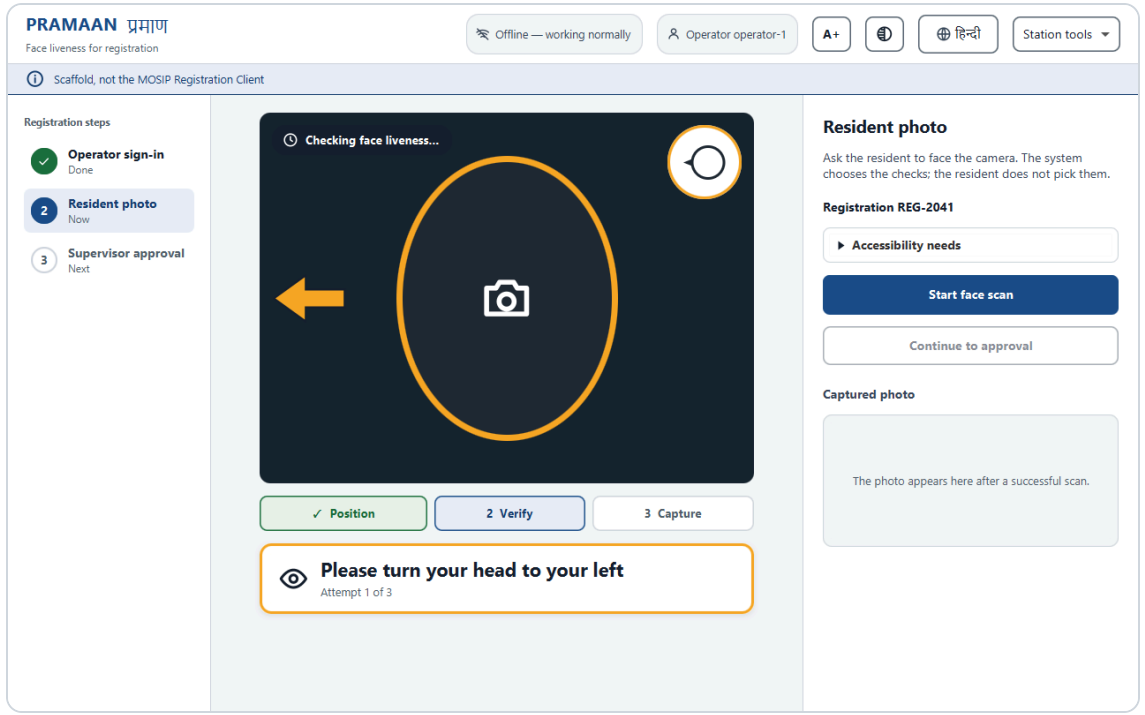
Blink, pulse and pupil reflex only ever vote *for* life. Hindi, icons, speech, accessibility profiles, infant mode, dual-control exceptions.

An SBI proposal

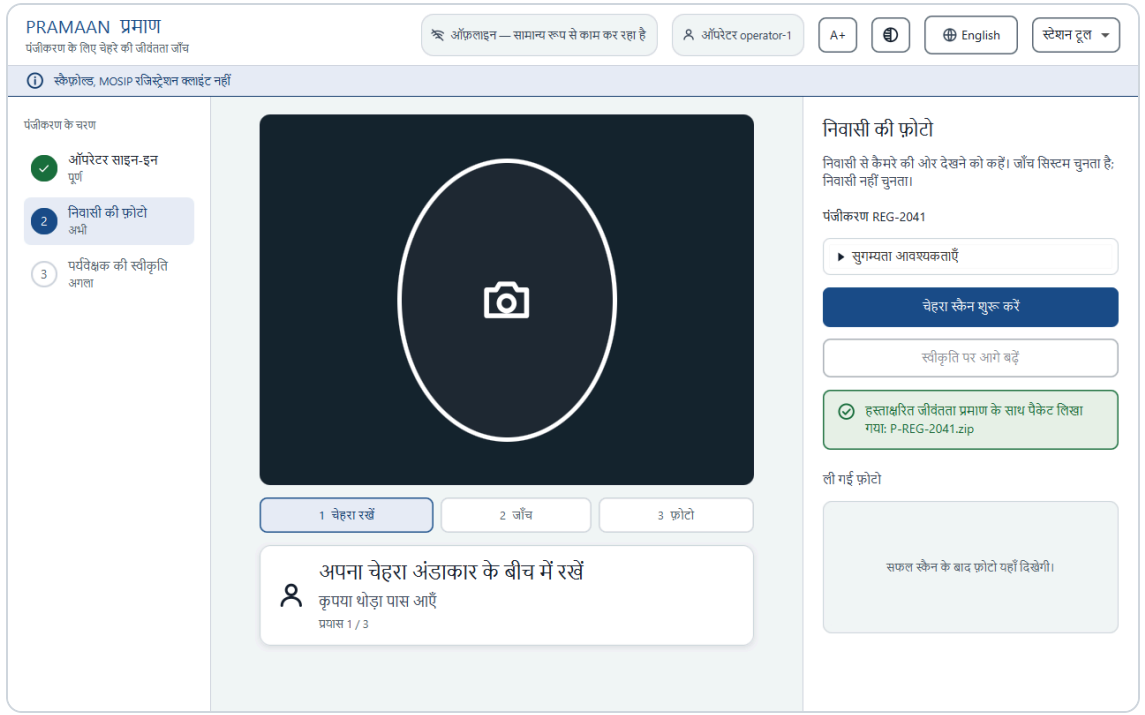
An optional Liveness Profile (signed preview, signed light log, signed burst) with a runnable reference device. Our own draft; not yet posted to MOSIP.

The station, as built

Rendered from the real code, offscreen. One plain sentence, an icon, no score. No live camera in these images.



Resident capture. Step rail, one instruction, Offline chip, the scaffold labelled as a scaffold.



Hindi, larger text, high contrast are one click. (Hindi drafted by machine, checked by the author.)

Fails closed. If the policy, the operating-point table or a model does not verify, nothing can be registered and the screen says what to check (*docs/media/fails-closed.png*). Failures show the resident one generic sentence and never the cause.

What is verified, and what is not

No real attack corpus exists, so no APCER or BPCER is claimed. Zero failures in 60 attacks would still allow a true rate near 6%.

Verified (one command each)

- **703 automated tests pass** (20 skipped on a fresh clone: no models)
- Flash code: 1,679,616 sequences = **20.7 bits**
- At most 3 colour changes a second, 900 sequences
- Insider signal: 6 in 40 at 2% gives z ≈ 5.87
- Transparency log matches the CT test vectors
- Each capture-swap attack is a failing test

Not done, not claimed

- No error rate on real attacks; thresholds provisional
- No full session on a phone or a certified SBI device (the models were timed on one phone)
- Desktop app is a scaffold, not the Registration Client
- No user has tried the screens; Hindi checked by the author only
- Flash, stereo, scene cues: synthetic data and fakes only

We attacked ourselves. On synthetic screen replays with the passive model assumed fooled and no flash light, the fusion accepted **4 of 85** display pitches; with the flash cue, **0**. Logged as R-03 with R-01 (moiré blind pitches); the fix (an accept needs a flash or stereo vote, else a challenge) is on for operators and supervisors, and not for residents.

Impact and next steps

Deployable on today's SBI, honest about what is unmeasured.

For MOSIP

- Closes the capture-swap gap with no spec change
- Leaves proof a server can re-check or re-score
- MPL-2.0; JavaFX 11 as in the Registration Client
- Inclusion rules keep it from locking residents out

What we would ask MOSIP

- How should an SBI extension be proposed?
- Licence terms of two anti-spoofing models?

Next, in order

1. **One real experiment, redone properly:** does the flash cue separate a live face from a replay on a laptop? (a first run was inconclusive: the replay was set up badly)
2. **A consented corpus**, then every threshold from data (calibrator built)
3. **A real device, a second camera, a phone**
4. **Real users:** operators and residents, an independent Hindi review

Repo: github.com/GargBhavya-tech/pramaan
Every claim and its command: docs/claim-sheet.md